Johanna Loyer

Postdoctoral Researcher in Cryptology and Quantum Computing

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Johanna Loyer

My research interests focus on the ***cryptanalysis*** *of* ***lattice****-based and* ***code****-based schemes, through the analysis of both classical and* ***quantum*** *attacks. I am particularly interested in transfers between these domains.*

Professional Experience

since Feb. 2025 — **Postdoctoral Researcher**, *Inria Saclay*, Palaiseau, GRACE team
with Thomas Debris-Alazard

Feb. 2024 – Jan. 2025 — **Postdoctoral Researcher**, *Centrum Wiskunde & Informatica (CWI)*, Amsterdam (Netherlands), Cryptology group
with Léo Ducas

Sept. 2020 – Dec. 2023 — **PhD in Computer Science**, *Inria*, Paris, COSMIQ team
Thesis title: *Quantum Cryptanalysis on Lattices and Codes*
advised by André Chailloux and Nicolas Sendrier

Teaching

2020–2023 — **Teaching Assistant**, *Sorbonne University*, Paris
- Tutorials: Quantum circuits and logic gates, Master 1 in Quantum Information
- Tutorials: Quantum information overview, Master 1 in Quantum Information
- Practical sessions: Python programming, Bachelor level, Year 1

June–August 2025 — **Internship supervision**, *3 months*, Samuel Godlewski, third-year Bachelor student in Applied Mathematics, Paris Dauphine University
Internship topic: Security analysis of a variant of the Wave digital signature scheme

Education

March–August 2020 — **Master's research internship**, *Inria*, Paris
Topic: Quantum sieving for lattices

2018–2020 — **Master's Degree in Applied Mathematics**, *University of Limoges*,
Master's thesis: Quantum enumeration for the Shortest Vector Problem

2017–2018 — **First year of Engineering School in Cybersecurity**, *INSA Centre Val de Loire*, Bourges

Publications

2026 **IACR Communications in Cryptology**, *Quantum security analysis of Wave*, Johanna Loyer
https://doi.org/10.62056/ae89ksuc2

2025 **CRYPTO**, *Wagner's Algorithm Provably Runs in Subexponential Time for SIS$^\infty$*, Léo Ducas, Lynn Engelberts, Johanna Loyer
https://doi.org/10.1007/978-3-032-01855-7_12

2025 **IACR Communications in Cryptology**, *Lattice Reduction via Dense Sublattices : A Cryptanalytic No-Go*, Léo Ducas, Johanna Loyer
https://doi.org/10.62056/ae0fhsfg

2025 **Design, Codes and Cryptography**, *Quantum sieving for code-based cryptanalysis and its limitations for ISD*, Lynn Engelberts, Simona Etinski, Johanna Loyer
https://doi.org/10.1007/s10623-024-01545-0

2023 **Post-Quantum Cryptography**, *Classical and Quantum 3 and 4-Sieves to Solve SVP with Low Memory*, André Chailloux, Johanna Loyer
https://doi.org/10.1007/978-3-031-40003-2_9

2021 **ASIACRYPT**, *Lattice Sieving via Quantum Random Walks*, André Chailloux, Johanna Loyer
https://doi.org/10.1007/978-3-030-92068-5_3

Submitted paper

2025 **Preprint**, *A Tight Quantum Algorithm for Multiple Collision Search*, Xavier Bonnetain, Johanna Loyer, André Schrottenloher, Yixin Shen
https://eprint.iacr.org/2025/1690.pdf

Specification document

2023 **Wave digital signature scheme**, *submitted to the NIST standardization process*, with G. Banegas, K. Carrier, A. Chailloux, A. Couvreur, T. Debris-Alazard, P. Gaborit, P. Karpman, R. Niederhagen, N. Sendrier, B. Smith, J-P. Tillich.
Document de spécification : https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-1/spec-files/wave-spec-web.pdf
Page officielle de Wave : https://tdalazard.io/wave.html

Ph.D. Thesis Manuscript

2023 **Quantum Cryptanalysis on Lattices and Codes**, *Inria Paris*, advised by André Chailloux and Nicolas Sendrier
https://theses.hal.science/tel-04390173/file/LOYER_Johanna_these_2023.pdf

Academic activities

2025 **Program Committee of PKC'26**, *https://pkc.iacr.org/2026/*
External reviewer, *for Eurocrypt 2023 et 2025, ANTS 2024, CRYPTO 2025*

Scientific Outreach

2025 **Member of EclairSI association**, Association of scientists contributing to informed public debate on computer security

2024 **Interview in the popular science magazine *Epsiloon* (n°38, July 2024)**
`https://www.epsiloon.com/tous-les-numeros/n38/panique_chez_les_cryptologues_post-quantiques/`

2023 **Organization of a outreach event**, *50 participants*, at the end of my Ph.D., aimed at raising awareness of cybersecurity, privacy, and the scientific method.
`https://johanna-loyer.github.io/page-personnelle/vulga/these.pdf`

2021 **Participation in RJMI**, Animation of speed-meeting sessions with high school girls to promote scientific careers.
`https://filles-et-maths.fr/rjmi/`

2018–2020 **Outreach talks in middle and high schools on cryptography**
focusing on the use of prime numbers in cryptography
`https://johanna-loyer.github.io/page-personnelle/vulga/RSA.pdf`

2018–2019 **Coordination of outreach talks on mathematics and post-high school orientation**, Conception and animation of sessions for 10th-grade classes on concrete applications of mathematics.
`https://johanna-loyer.github.io/page-personnelle/vulga/maths.pdf`

2018 **Playful workshops in robotics and programming (Scratch, Ozobot)**, in partnership with the INSPE of Bourges, for primary school students.

Languages

Français	native	Python	Expert
Anglais	fluent	SageMath	Expert
Norvégien	intermediate	C	Advanced